Assembly Guide

This guide explains how I assembled the mechanical structure of my custom 6-DOF robotic arm. The parts were designed in Onshape, 3D printed, and assembled with hobby servos, screws

Hardware

- All parts were described in further detail in the Parts List in the GitHub repository

Before Assembly

1. Remove support material from the printed parts.
2. Clean the screw holes and servo openings.
3. Use a small file or sandpaper if a part does not fit.
4. Check each part against the Onshape model.
5. Test-fit the parts without screws first.
6. Make sure moving joints can rotate without rubbing.

Assembly

1. Finish prep steps above for all parts
2. Mount each servo in its matching printed location as seen on the CAD
3. Assemble the arm from the base upward, attaching the arm sections, wrist components, and gripper
4. Install servo horns and linkage components of the gripper leaving enough slack for the joints to move
5. Route the servo wires along the arm, leaving enough slack for each joint to move freely
6. Check for interference, pinched wires, or loose screws before moving on to code

Leaving the screws slightly loose during the first assembly makes it easier to align the parts. Fully tighten the hardware after checking that each joint moves correctly.

Servo Horn Alignment

Servo horns should be installed after the servos have been moved to their neutral position in the control code.

The general process is:

1. Temporarily install the servo without permanently attaching the horn.
2. Use the test code to move the servo to its center position.
3. Turn off power.
4. Place the horn in the closest mechanically centered position.
5. Install the horn screw.
6. Attach the linkage or printed joint.
7. Test the movement using a small range.

The neutral position for this project is approximately 90 degrees, but the physical “straight” position may be different for each joint because the servos are mounted in different orientations.

Wire Routing

Route the servo wires so they:

- Do not pass through screw holes.
- Do not get pinched by moving parts.
- Do not rub against sharp printed edges.
- Have enough slack for the joint to move.
- Stay away from servo horns and gears.
- Can be removed later for repairs.

Use small cable ties, adhesive clips, or printed cable guides where appropriate. Do not make the wires so tight that they pull on the servo connectors when the arm moves.